

Semester	I	L	T	P	C	COURSE CODE
Regulation	V26	0	0	3	1.5	
Name of the Course	AI Tools and Applications					
Branch	Common to all branches					

Syllabus

Course Outcomes:

After successful completion of the course, the student will be able to:

CO	Course Outcome	Knowledge Level
CO1	Explain AI concepts and identify suitable AI tools for different tasks.	K2
CO2	Apply prompting techniques to obtain useful and accurate AI-generated responses.	K3
CO3	Experiment with AI tools for data analysis, visualization, and interpretation.	K3
CO4	Apply AI tools to coding, writing, presentations, and productivity tasks.	K3
CO5	Illustrate ethical, privacy, reliability, and security issues in using AI tools.	K2

Suggested Tools: Conversational and text-generation tools (e.g., ChatGPT, Gemini, Claude); AI coding assistants (e.g., GitHub Copilot, Gemini Code Assist); AI data-analysis tools (e.g., ChatGPT, Gemini, Microsoft Copilot); AI image-generation tools (e.g., Adobe Firefly, Microsoft Designer); and AI writing and productivity tools (e.g., Grammarly, Microsoft Copilot).

Unit I: Introduction to AI and the AI Tools Landscape

1. What is AI, Machine Learning, and Generative AI
2. Core categories of AI tools: conversational/text, code assistants, data-analysis tools, image-generation tools, and writing/productivity tools.
3. How large language models process and generate responses (conceptual, non-mathematical overview).
4. Criteria for selecting the right AI tool for a given task — accuracy needs, cost, data sensitivity, and output format.
5. Setting up accounts and workspaces; understanding free-tier versus paid-tier capabilities and limitations.

Exercise 1:

1. Create accounts on two AI platforms and compare their responses to a simple engineering question (e.g., "explain the difference between analysis and design").
2. Prepare a one-page comparison chart of five AI tools by category and primary use case.
3. Write a short reflection identifying one task in your daily academic life where an AI tool could help.

Unit II: Prompt Engineering and Generative AI Tools

1. Anatomy of a good prompt — context, clear instruction, desired format, and constraints.
2. Zero-shot, few-shot, and Structured reasoning and problem decomposition and when to use each.
3. Role-based and context-setting prompts (e.g., asking the AI to respond as a reviewer, tutor, or domain expert).
4. Iterative prompt refinement — diagnosing a poor AI response and rewriting the prompt to fix it.
5. Using generative AI for brainstorming, summarizing, and drafting technical content.

Exercise 2:

1. Write three versions of a prompt (basic, detailed, role-based) for the same task and compare outputs.
2. Use an AI tool to summarize a one-page technical article and verify the summary against the original.
3. Use structured prompting to solve a multi-step numerical or logical problem and verify the result independently.

Unit 3: AI Tools for Data Analysis and Visualization

1. AI-assisted spreadsheet and dataset analysis using natural-language commands.
2. Natural-language querying of datasets to extract summary statistics and insights.
3. AI tools for generating charts, graphs, and visual summaries from tabular data.
4. Interpreting AI-generated statistical summaries correctly and recognizing their limitations.
5. Identifying and correcting errors or misinterpretations in AI-generated data analysis.

Exercise 3:

1. Use an AI tool to clean and summarize a small sample dataset (e.g., class attendance or marks data).
2. Generate three different chart types from the same dataset using AI prompts and justify which is most appropriate.
3. Cross-check one AI-generated statistical claim manually and report any discrepancy found.

Unit 4: AI Tools for Coding, Writing, and Productivity

1. AI coding assistants — code generation, explanation, and debugging support for simple programs.
2. AI writing assistants for reports, emails, and technical documentation.
3. AI tools for scheduling, note-taking, and meeting/lecture summarization.

4. AI-assisted presentation and slide-outline generation from a topic prompt.
5. Integrating AI tools into an efficient day-to-day academic and engineering workflow.

Exercise 4:

1. Use an AI coding tool to write and debug a simple program for a given problem statement.
2. Draft a formal email or short report using an AI writing assistant, then manually edit it for accuracy and tone.
3. Generate a slide outline for a 5-minute seminar topic using an AI tool and refine it.

Unit 5: Ethics, Reliability, and Emerging Trends

1. Bias and hallucination in AI outputs, and techniques for detecting and fact-checking them.
2. Academic integrity and appropriate citation of AI assistance in coursework and submissions.
3. Data privacy and security considerations, including obligations under India's Digital Personal Data Protection Act, 2023 (DPDP Act), when uploading personal or institutional data.
4. Copyright and intellectual property considerations in AI-generated text and image content.
5. Emerging trends — multimodal AI and AI agents — and their implications for future engineering practice.

Exercise 5:

1. Identify and document a hallucination or bias instance produced by an AI tool on a factual query.
2. Draft an "AI usage declaration" for a hypothetical assignment, citing tools and extent of use appropriately.
3. Review the data-handling/privacy policy of one AI tool used in this course and identify one DPDP Act, 2023 compliance consideration relevant to student data.

Text Books:

1. Generative AI in practice: 100+ amazing ways generative artificial intelligence is changing business and society, Marr, B. (2024), Wiley.
2. Co-intelligence: Living and working with AI, Mollick, E. (2024), Portfolio/Penguin.

References:

1. Prompt engineering for generative AI: Future-proof inputs for reliable AI outputs. Phoenix, J., & Taylor, M. (2024), O'Reilly Media.
2. Introduction to generative AI: A practical guide to generative AI, large language models, and more, Dhamani, N., & Engler, M. (2024), Manning Publications.
3. Education and technology: Key issues and debates, Selwyn, N. (2022), 3rd ed. Bloomsbury Academic.